Controller Design via Root Locus — Influence of High Dynamics

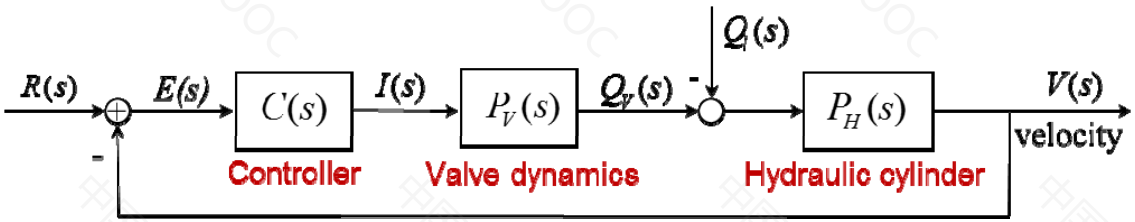
(基于根轨迹的控制器设计—高频动力学影响)

Ex.RL3 Motion Control of Hydraulic Cylinders

The engineers implemented the PD controller on the actual hydraulic excavators but found out that the resulting closed-loop system still exhibits severe chattering problem and, even worse, the closed-loop system could become unstable sometimes. To explain this phenomenon, let us carefully examine the hardware used to implement the controller. The flow rate to the cylinder is indirectly regulated by a set of valves. In the previous design, the valve dynamics were neglected and the flow rate was considered to be the control input although actual control input should be the current (or the force) to the valve used to open or close the valve. However, due to the low-end valves used in the actual hydraulic excavator, the responses of the valves are not very fast and the relationship between the actual flow rate and the current to the valve is described by a second-order system with a natural frequency of 7Hz and a damping ratio of 0.7. You are required to analyze this actual system and explain why the PD controller designed before does not work for this system. **(工程师将PD控制器作用于实际液压挖掘机后，结果发现闭环系统仍显示出严重的振荡问题。而且，闭环系统有时会不稳定。为解释这个现象，让我们仔细检查控制系统的硬件。液压缸流量由一套阀来非直接调节。在之前的设计中，忽略了阀动力学，认为控制输入是流量。然而，实际控制输入是作用在阀上的电流或力，用于打开或关闭阀。由于实际液压挖掘机上用的是低端阀，阀的响应并不很快，其实际流量和阀电流之间的关系由一个自然频率是7Hz、阻尼比是0.7的二阶系统来描述。要求你来分析实际系统并解释为什么之前设计的PD控制器对这个系统没有效。)**

Ex.RL3 Motion Control of Hydraulic Cylinders

***Block Diagram of Feedback Control System*:**



$$P_H(s) = \frac{71}{s^2 + 3.77s + 355} = \frac{71}{(s - p_{p1})(s - p_{p2})}$$

$$P_V(s) = \frac{(7 \times 2\pi)^2}{s^2 + 2 \times 0.7 \times (7 \times 2\pi)s + (7 \times 2\pi)^2} = \frac{1934}{(s - p_{p3})(s - p_{p4})}$$

$$G_p(s) = G_H(s)G_V(s) = \frac{K_p N_p(s)}{D_p(s)}$$

where, $K_p = 1934 \times 71 = 137314, N_p(s) = 1, D_p(s) = (s - p_1)(s - p_2)(s - p_3)(s - p_4)$

$$p_{p1,2} = -1.89 \pm 18.76j$$

$$p_{p3,4} = -30.8 \pm 31.4j$$

Ex.RL3 Motion Control of Hydraulic Cylinders

PD feedback controller designed before:

$$C(s) = 0.45s + 1.59 = \underbrace{0.45}_{K_c}\ \underbrace{(s+3.5)}_{N_c(s)}$$

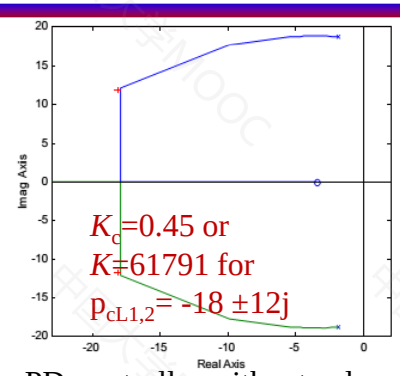
$$p_{p1,2} = -1.89 \pm 18.76j$$

$$p_{p3,4} = -30.8 \pm 31.4j$$

Open-Loop Poles and Zeros: $D(s) = 0 \Rightarrow$ four OL poles at p_1, p_2, p_3, p_4.

$N(s) = 0 \Rightarrow$ one OL zero at $z_1 = -3.5$

Two extra open-loop poles $p_{3,4}$ changes the root-locus dramatically since $p_{3,4}$ are close to the desired CL poles at -18±12*j* (两个开环极点$p_{3,4}$明显改变了根轨迹，因为$p_{3,4}$靠近期望的闭环极点)



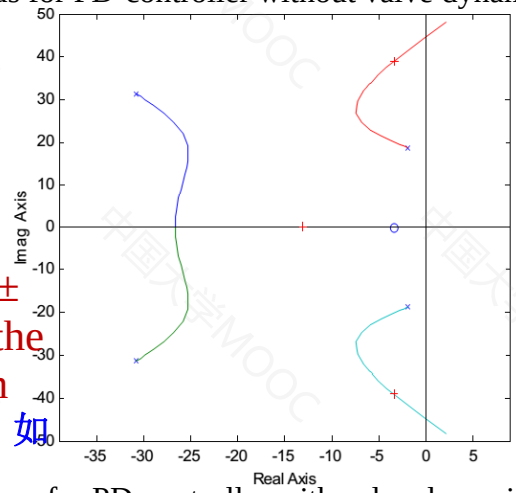
Root Locus for PD-controller without valve dynamics

The resulting CL poles: $p_{CL1,2} = -3.7 \pm 38j,\ p_{CL3} = -13.8,\ p_{CL4} = -44$

The resulting CLTF: $$T(s) = \frac{61791(s+3.5)}{s^4 + 65s^3 + 2522s^2 + 90943s + 9.03\times10^5}$$

Q: Can you explain the phenomenon that the engineer encountered (你能解释工程师面临的现象吗?)

Yes. Since the four CL poles have a pair of lightly damped poles at –3.7 ± 38*j*, the response is very oscillatory. As seen, if we increase the gain K_c, the two lightly amped poles may become unstable and destabilize the system (是的。因为四个闭环极点有一对欠阻尼极点**–3.7 ± 38*j***，因此响应非常振荡。如果我们增大增益K_c,这两个欠阻尼极点可能变得不稳定并使系统不稳定)。



Root Locus for PD-controller with valve dynamics

Ex.RL3 Motion Control of Hydraulic Cylinders

Method 1: Feedback controller with stable pole/zero cancellation **(稳定零极点相消反馈控制器):**

Use stable pole/zero cancellation to raise the valve bandwidth

Add compensator to raise the valve bandwidth to a natural frequency of ω_f=20Hz=20×(2π) rad/s and a damping ratio of ζ_f=0.8 **(增加补偿器来提高阀带宽至自然频率ω_f=20×(2π) rad/s和阻尼比ζ_f=0.8)**:

$$C_f(s) = \frac{\omega_f^2}{\omega_v^2}\frac{s^2+2\zeta_v\omega_v s+\omega_v^2}{s^2+2\zeta_f\omega_f s+\omega_f^2} = 8.16\frac{(s-z_{c2})(s-z_{c3})}{(s-p_{c1})(s-p_{c2})}$$

added zeros to cancel the poles of valve dynamics are **(增加零点来消除阀动态的极点)**

$$z_{c2,3} = p_{p3,4} = -30.8 \pm 31.4j$$

and the added new poles have $\omega_f = 20Hz$, $\zeta_f = 0.8$ as required **(增加满足ω_f, ζ_f要求的新极点)**

$$p_{c1,2} = -100.5 \pm 75.4j$$

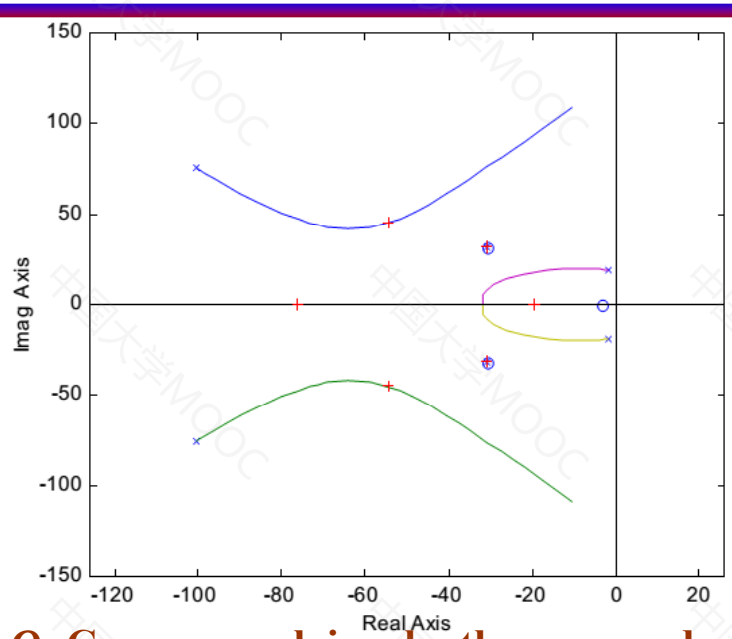
Overall Controller $C(s) = 0.45(s+3.5)\,C_f(s) = 3.67\dfrac{(s+3.5)\,(s-z_{c2})\,(s-z_{c3})}{(s-p_{c1})\,(s-p_{c2})}$

Open-Loop Poles and Zeros **(开环极点和零点)**:

$D(s) = 0 \Rightarrow$ six open-loop poles at $p_1, p_2, p_3, p_4, p_5, p_6$.

$N(s) = 0 \Rightarrow$ three open-loop zero at z_1, z_2, z_3

Ex.RL3 Motion Control of Hydraulic Cylinders



Open-loop poles and zeros (开环极点和零点):

z_1=-3.5, $z_{2,3}$=-30.8 ± 31.4j,

$P_{1,2}$=-1.89 ±18.76j,

$p_{3,4}$= = -30.8 ± 31.4j

$p_{5,6}$=-100.5 ±75.4j

The six CL poles using the controller are:
(采用控制器后系统的6个闭环极点)

$$-19.7,\ -74.7,$$

$$-30.8 \pm 31.4j$$

$$-55.2 \pm 44.3j$$

***Q:* Can you explain why the proposed controller works ? (你能解释为什么所提出的控制器有效)**

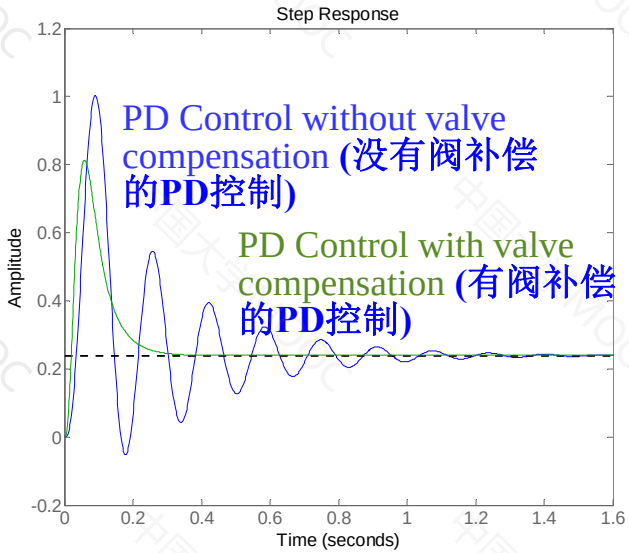
The six CL poles using the controller C(s) are -19.7, -74.7, -30.8 ±31.4 j, -55.2 ±44.3 j which are all stable and well damped (6个闭环极点是稳定的且阻尼较适中)

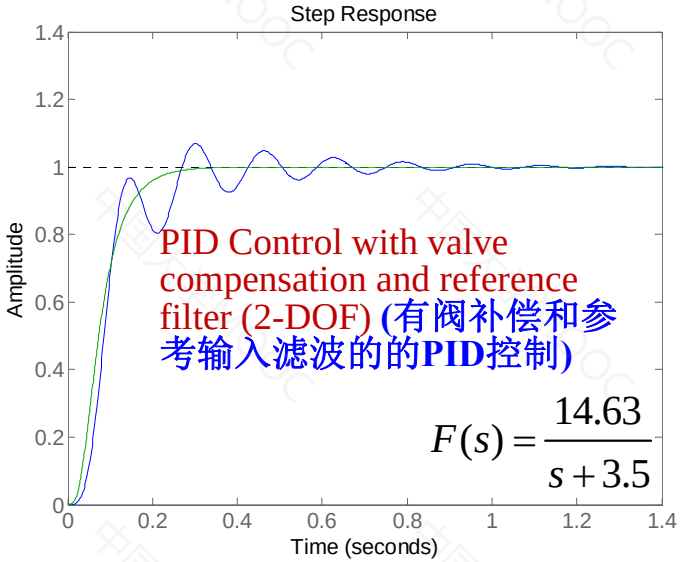
The resulting CLTF is:

$$T(s) = \frac{5\times10^5(s+3.5)(s-z_{c2})(s-z_{c3})}{s^6+266.4s^5+3.15\times10^4 s^4+2.1\times10^6 s^3+7.9\times10^7 s^2+1.7\times10^9 s+1.4\times10^{10}}$$

Ex.RL3 Motion Control of Hydraulic Cylinders

Simulation





Large overshoot due to the slow CL zero at -3.5 can be removed by using 2-DOF controller and a reference filter TF with a pole at -3.5 (因慢的闭环I零点导致大的超调能通过两自由度控制器和设置极点在-3.5的参考滤波传递函数来消掉)

Q: When can you use pole/zero cancellation technique? (我们什么时候能用极点/零点相消技术？)

Note that the cancelled poles remain as closed-loop poles. This indicates that we can only use the technique to cancel stable open-loop poles; those poles should be away from imaginary axis sufficiently (注意消去的极点仍然是闭环系统的极点，这表明我们仅能使用此技术来消去稳定的开环极点，这些极点应当离虚轴足够远).

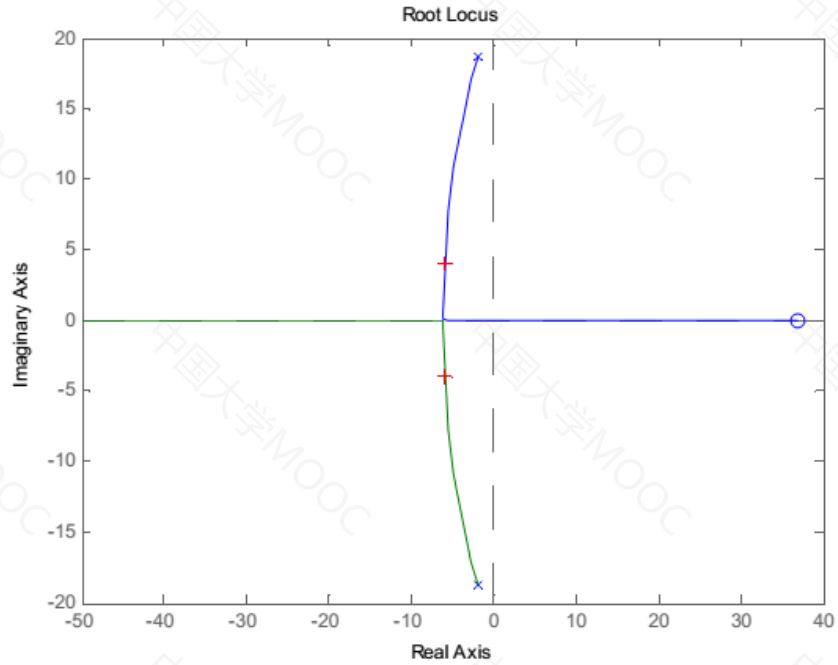
Ex.RL3 Motion Control of Hydraulic Cylinders

Method 2: Limit the CL Bandwidth so that valve dynamics can be ignored **(限制闭环带宽使阀动态能被忽略)**

Consider the following PD controller which would place the two CL poles at -6 ±4*j* when the valve dynamics are not considered **(当阀动力学没有考虑时，考虑以下PD控制器，其配置的两个闭环极点在-6 ±4*j*):**

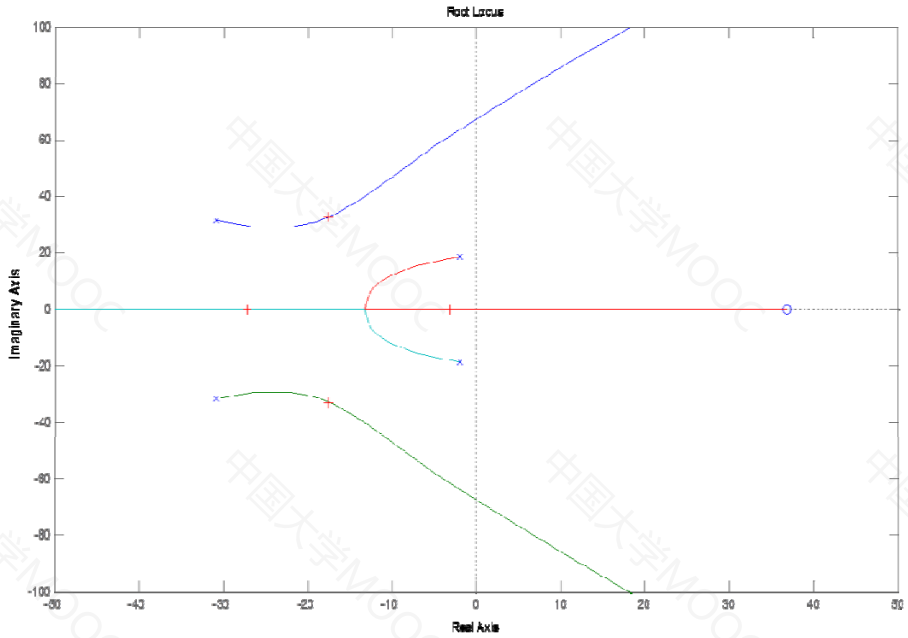
$$C(s) = 0.116s - 4.27 = 0.116(s - 36.85)$$

Noting that this PD controller is quite unusual in the sense that P-gain is negative! **(注意PD控制器很不寻常，因为P增益是负的)**



Ex.RL3 Motion Control of Hydraulic Cylinders

With the valve dynamics, the root-locus of the CL system with a PD controller of the form $C(s)=K_{CD}(s\text{-}36.85)$ is shown below for $K_{CD}>0$ **(有阀动力学时，带PD控制器C(s)= $K_{CD}(s\text{-}36.85)$的闭环系统的根轨迹显示如图)**



With K_{CD}=0.116, the resulting CL poles are

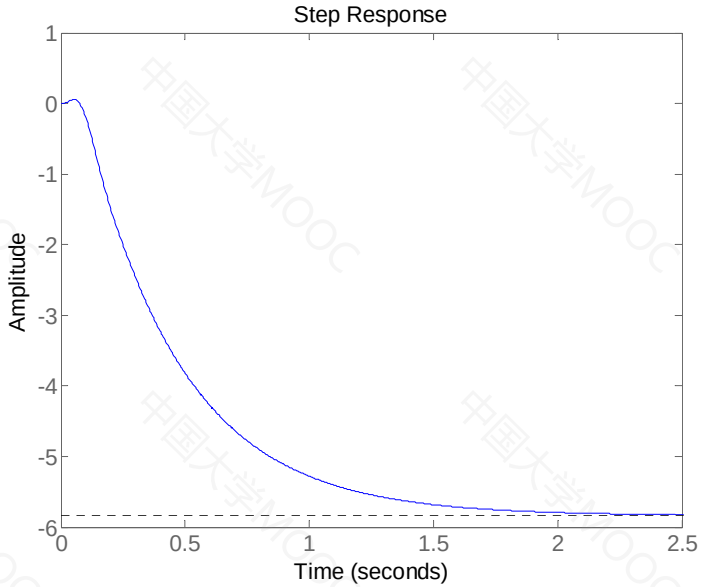
$$-17.31 \pm j32.95, \quad -28.15, \quad -2.58$$

Ex.RL3 Motion Control of Hydraulic Cylinders

And the CLTF is

$$T(s)=\frac{15920s-586700}{s^4+65.35s^3+2522s^2+45090s+100600}$$

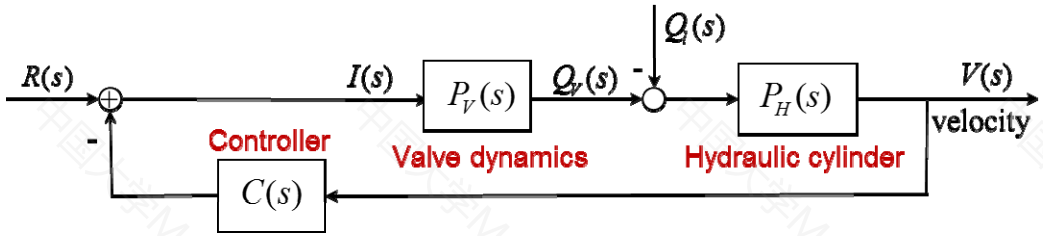
which has a unit step response shown below



revealing no oscillating responses but exhibiting undershoot phenomenon due to the unstable zero appeared in the CLTF (没有振荡，但是因闭环传递函数存在不稳定零点而产生负调现象).

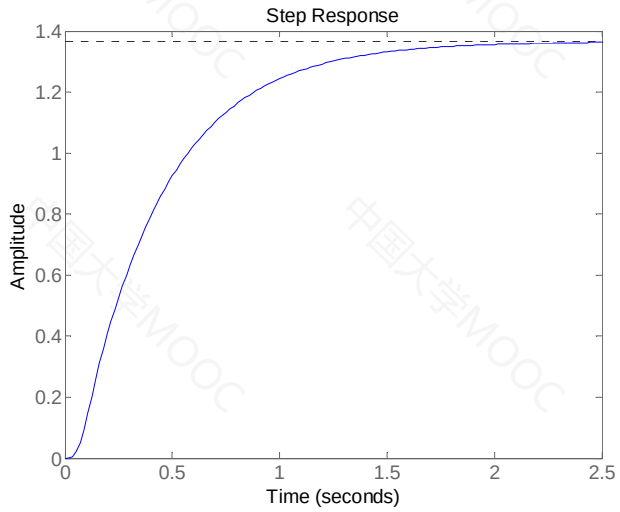
Ex.RL3 Motion Control of Hydraulic Cylinders

The undershoot problem can be avoided by placing the PD controller in the feedback path instead of in the forward path: **(把PD控制器放置在反馈通路上而不是前向通路上可解决负调问题)**



CLTF: $$T(s)=\frac{137314}{s^4+65.35s^3+2522s^2+45090s+100600}$$

Unit Step Response:



Summary

- **Motion Control of Hydraulic Cylinders** (液压缸的运动控制)
  - ✓ **Without valve dynamics** (无阀动力学)
  - ✓ **With valve dynamic** (有阀动力学)